A Conceptual Primer on World-Model Markov Reasoning, Hidden-State Inference, and POMDP Belief Semantics

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Abstract

This note is a conceptual primer for a research line connecting world-model calculus, Markov semantics, higher-order Markov chains, hidden Markov models, credal uncertainty, and POMDP-style reasoning. Its purpose is not to archive implementation history but to preserve the strongest conceptual picture now available so that future technical work and future conversations can resume from a stable basis.

The central thesis is that probabilistic symbolic reasoning becomes reliable only when it sharply separates five layers that are often conflated:

1. the explicit semantic model that determines probabilities,
2. the evidence or posterior state carried by a reasoning system,
3. the query surface exposed to users,
4. the operational control layer that decides what computation to perform,
5. and the higher-order uncertainty induced by hidden structure, non-identifiability, or model ambiguity.

On this view, additive evidence revision remains a mathematically privileged fragment, but it is not the universal semantics of hidden-state reasoning. For ordinary finite-state Markov chains, predictive path semantics can be exposed honestly through world-model queries. For higher-order Markov chains, the right reduction is to first-order chains on context states. For hidden Markov models, complete-data evidence remains additive, but observed-only belief tracking is fundamentally non-additive; the correct response is filtering, smoothing, and, in the ambiguous case, credal or interval-valued belief representations. Action-conditioned hidden Markov models then provide a clean structured entry point for POMDP-capable world-model reasoning.

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1 Purpose and scope

This document consolidates a cluster of ideas that belong together:

- world-model semantics as the truthmaking layer for probabilistic logic;
- honest Markov path semantics in place of ad hoc multi-step chaining;
- higher-order Markov semantics via context-state reduction;
- hidden-state inference via forward, backward, filtering, and smoothing;
- observed-only non-identifiability and its implications for additive evidence calculi;
- credal world-model views as an honest response to ambiguity;
- and controlled hidden Markov models as a bridge into POMDP-style reasoning.

The note is strictly conceptual. It is written to be useful both as an academic synthesis and as a primer for future technical work. It therefore emphasizes conceptual boundaries, canonical constructions, and theorem-shaped open problems rather than implementation detail.

2 The guiding separation: semantics, evidence, queries, and control

2.1 The main distinction

The first indispensable distinction is:

> The semantic model determines what is true; the carried evidence or posterior state records what a reasoner knows; the query surface exposes selected views of that state; and the control layer decides what computation to perform next.

Many failures of probabilistic symbolic reasoning arise from collapsing these roles into one object.

- A probability in a semantic model is not the same thing as a confidence attached to a derived symbolic judgment.
- A local truth value is not automatically a reusable semantic fact.
- A search-control score is not the same thing as posterior support.
- A posterior belief state is not the same thing as an additive multiset of observations unless the underlying model class genuinely supports such a reduction.

2.2 The additive fragment remains special

The additive world-model fragment should not be discarded. It remains canonical for families where independent observations aggregate through sufficient statistics. In that setting, evidence accumulation is literally additive. The point is not that additivity is wrong; the point is that it is correct only in the fragment where the semantics really supports it.

Principle 1 (Additive privilege)**.** *When a model class admits a sufficient-statistic posterior that aggregates by addition, the additive world-model semantics is the canonical surface for that class.*

Positive example:

- Beta–Bernoulli and Dirichlet–multinomial families aggregate evidence by counts, so additive evidence states are exactly right.

Negative example:

- Observed-only hidden-state inference generally does not reduce to an additive multiset statistic that recovers the correct posterior over latent states.

3 From local chaining to explicit predictive semantics

3.1 Why naive probabilistic chaining fails

The tempting but unreliable idea is that one can propagate local truth values through arbitrary chains as if local support values composed like logical probabilities. This fails for familiar reasons:

- hidden overlap and dependence are forgotten,
- derived links are reused as if they were independent raw premises,
- and multi-step composition is forced into an additive or deductive form that the semantics does not justify.

3.2 What survives

What survives is the narrower but much cleaner claim:

> Multi-step probabilistic reasoning should be exposed through direct semantic queries against an explicit model, not through generic reuse of locally derived support values.

For finite-state Markov chains, this means that multi-step queries should be read as *predictive path probabilities*. The right semantics is sequential posterior or predictive composition, not generic additive deduction.

4 Finite-state Markov world-model semantics

4.1 Single-step queries

The simplest Markov query asks for the probability of a one-step transition

$$P(X_{t+1} = x' \mid H_t),$$

where H_t is the relevant history summary. In a conjugate finite-state setting, the sufficient statistic can be read off from row counts and boundary data.

4.2 Path queries

The conceptual improvement over naive chaining is to introduce direct path queries

$$P(X_{t+1} = x_1, \ldots, X_{t+n} = x_n \mid H_t).$$

These are not mere chains of local truth values; they are model-defined path probabilities.

Principle 2 (Predictive over deductive)**.** *For finite-state Markov reasoning, multi-step query strength should be defined by predictive path semantics, not by additive reuse of locally derived transition judgments.*

4.3 Threshold semantics

Once direct predictive query strength exists, threshold theorems become meaningful: one may ask whether a path or transition has support above a given cutoff, but this thresholding happens *after* semantic evaluation, not in place of it.

5 Higher-order Markov chains through context-state reduction

5.1 The basic idea

An order-m Markov chain on symbols can be reinterpreted as a first-order Markov chain on length-m contexts. This is not merely a coding trick. It is the mathematically right reduction for bringing higher-order chains into the same finite-state kernel and world-model infrastructure.

Definition 1 (Context-state reduction)**.** *An order-m chain on symbols x_t is represented as a first-order chain on states*

$$c_t = (x_t, \dots, x_{t+m-1}),$$

with transitions given by shift-consistent context extension.

5.2 Why this matters

This reduction gives three conceptual payoffs.

1. It turns higher-order sequence laws into ordinary first-order path laws on context states.
2. It identifies the right sufficient statistic: $(m + 1)$-gram counts.
3. It makes higher-order reasoning compatible with the same predictive and evidence-carrying surfaces used for first-order finite-state chains.

5.3 Raw symbol semantics versus context semantics

There are two distinct but equivalent views:

- context-trajectory semantics, where the underlying process lives on contexts;
- raw-symbol semantics, where one projects context trajectories back down to symbol sequences.

The raw-symbol cylinder law is conceptually important because it shows that the context-state reduction is not just an auxiliary proof device; it is genuinely equivalent to the higher-order symbolic process.

6 Hidden Markov models: from complete data to observed-only inference

6.1 The two views of an HMM

A finite HMM has:

- a latent Markov chain,
- an emission kernel,

- and an induced observed process obtained by summing or integrating over latent paths.

From the complete-data point of view, the model remains additive in a useful way: one can accumulate latent transition counts and emission counts. From the observed-only point of view, however, the correct object is not an additive count summary but a posterior distribution over latent states.

6.2 Forward and backward semantics

The forward message computes the unnormalized mass of ending in latent state x after an observed prefix. The backward message computes the continuation likelihood of the future observations given latent state x. Their product recovers the standard smoothing factorization.

Conceptually, four objects matter:

- observed-word probability,
- filtering posterior,
- smoothing posterior,
- and two-slice latent posterior.

6.3 Filtering

Filtering is the posterior belief state

$$\mathrm{Bel}_t(x) = P(X_t = x \mid Y_{\leq t}, A_{\leq t}),$$

or its action-free specialization when no actions are present. This is the load-bearing object for observed-only reasoning.

6.4 Smoothing

Smoothing is the two-sided posterior

$$P(X_t = x \mid Y_{\leq T}, A_{\leq T-1}),$$

or its finite-prefix analogue. The conceptual lesson is that smoothing is not an optional embellishment. It makes explicit that observed-only reasoning is posterior reasoning over hidden trajectories, not a direct additive tally.

7 The non-identifiability boundary

7.1 What the obstruction says

The central negative insight is:

> Observed-only hidden-state belief cannot in general be recovered from an additive multiset-based evidence surface over observations alone.

The cleanest witness is a label-swap construction: two hidden models can induce the same observed distribution while assigning incompatible latent posteriors. This is not a pathology. It is the structural reason observed-only HMM belief tracking cannot simply inherit the complete-data additive world-model infrastructure.

7.2 Why the result matters

This obstruction has three consequences.

1. It marks the boundary of additive WM-PLN reasoning for hidden-state models.
2. It justifies filtering and smoothing as first-class semantic objects.
3. It motivates credal or interval-valued belief representations when the observations underdetermine latent structure.

8 Credal world-model semantics for observed-only ambiguity

8.1 The conceptual move

When the observations do not identify a unique hidden model, the right response is not to pretend that a single posterior strength exists. Instead one should carry a family of compatible hidden models and expose lower and upper support for a latent query.

Definition 2 (Credal filtering truth value)**.** *Given a finite family of compatible hidden models Θ, an observed trace z, and a latent query q, define*

$$\underline{s}(q \mid z) = \inf_{\theta \in \Theta} P_\theta(q \mid z), \qquad \overline{s}(q \mid z) = \sup_{\theta \in \Theta} P_\theta(q \mid z).$$

This interval is the honest observed-only world-model answer when latent structure is ambiguous.

8.2 Why this is conceptually natural

The credal move is not a retreat from semantics. It is the semantics of ambiguity. In this sense, the obstruction theorem is not merely a negative result. It says what the next representation should be.

Positive example:

- a label-swapped HMM family can induce the full interval $[0, 1]$ for a latent-state query after a singleton observation.

Negative example:

- the interval need not collapse even for very short traces, so observed-only ambiguity is not a rare asymptotic phenomenon.

9 Controlled HMMs as a structured bridge to POMDP reasoning

9.1 Why action-conditioning matters

A POMDP is not merely an HMM with observations. It is an action-conditioned partially observable process. The clean structured bridge is therefore a controlled finite HMM:

- latent prior,
- action-conditioned latent transition kernel,
- emission kernel,
- filtering and smoothing on completed action-observation cycles.

9.2 What this unlocks

This does *not* create the general agent framework from scratch. Rather, it provides a structured finite-state instance of an already general history-conditional environment formalism. The main gain is conceptual:

> The belief state of a finite POMDP can now be expressed as a controlled filtering posterior in the world-model reasoning layer.

9.3 The controlled Solomonoff leg

The controlled story now has a conservative universal-prediction leg as well. A controlled finite HMM induces a controlled prefix law on completed action-observation traces, and along any fixed action stream one can condition that law down to an ordinary observation-prefix measure. The ordinary finite-alphabet Solomonoff semimeasure M_2 can then be lifted back into the controlled setting by ignoring actions and acting as a comparator on the observation side. This yields a real theorem:

> For any fixed action stream, if the induced observation law is lower semicomputable, then the controlled Solomonoff comparator satisfies the standard finite-horizon log-loss regret bound against that controlled hidden Markov law.

Positive example:

- the controlled Markov / Solomonoff / world-model triangle is now closed at the level of fixed-action-stream regret bounds.

Negative example:

- the current comparator is deliberately conservative: it is not yet a genuinely action-aware universal mixture.

9.4 A genuine computability boundary

The lower-semicomputability side condition in the controlled Solomonoff theorem is not a technical embarrassment. It reflects a real domain boundary. The raw controlled finite HMM parameter record allows arbitrary probability measures, so lower semicomputability does not follow automatically from the bare semantic object. The right response is to package a computable or lower-semicomputable subclass explicitly, not to erase the assumption by fiat.

9.5 The correct interpretation

In this context, "unlocking POMDP reasoning" means:

- one has a structured controlled latent-state model,
- one has a bridge from that model to a generic environment interface,
- and one can state planning semantics in terms of belief-state update.

What it does *not* mean is that additive evidence alone now suffices for hidden-state planning. The obstruction theorem still applies.

10 World-model reasoning for POMDPs

10.1 What proper WM reasoning should mean here

Proper world-model reasoning for POMDP-like systems should mean:

1. a semantic model with explicit latent and observed dynamics,
2. belief-state update equations,
3. query surfaces for filtering, smoothing, and prediction,
4. planning or control semantics that read those belief states correctly,
5. and an honest ambiguity layer when observations fail to identify a unique latent state model.

10.2 What it should not mean

It should not mean:

- reusing additive observation multisets as if they were sufficient for latent-state posteriors,
- calling any local truth-value propagation scheme "belief update",
- or hiding hidden-state ambiguity under a single scalar confidence.

11 The de Finetti and mixture story

11.1 What is trivial and what is not

Mixture theorems come in two sharply different forms.

Trivial direction. For each fixed parameter θ, the law induced by θ can always be represented as a Dirac mixture concentrated at θ.

Nontrivial direction. One wants a real mixture theorem: finite discrete mixtures, or ultimately arbitrary Borel mixtures, together with an honest image characterization of the laws thus obtained.

11.2 The naming lesson

The conceptual lesson is methodological:

> Dirac-witness existence theorems are legitimate, but they should be named as Dirac-witness theorems rather than as if they already established the full de Finetti image theorem.

This matters because the difference is mathematically substantive. A statement may be correct as written while still overselling what kind of factorization has actually been proved.

11.3 Higher-order and HMM mixture surfaces

For higher-order Markov models, the first honest nontrivial step is a finite discrete mixture theorem on context-state parameters. For hidden Markov models, the same applies to latent Markov parameters with fixed emission kernel. The genuinely hard open layer is the arbitrary-Borel image theorem in both settings.

12 The ProbMeTTa lesson: semantic mirror versus source semantics

12.1 The initial gap

Another general lesson emerging from this line of work is the distinction between:

- a semantic theorem downstream of a compilation or translation relation,
- and a formal semantics of the actual source-level runtime language.

It is easy to prove a theorem of the form:

> If the source system compiles to a certain semantic object, then the downstream theorem holds.

This is useful, but it is not the same as formalizing the actual source operators and proving the compiler relation from the runtime semantics.

12.2 The right target

The right target is not byte-for-byte replay of an implementation. The right target is a mathematically honest source semantics for the runtime core, together with adequacy theorems connecting it to the already-proved semantic backend.

This yields a more general methodological rule:

Principle 3 (Semantic fidelity over operational mimicry)**.** *When formalizing a symbolic runtime, the goal should be semantic adequacy of the source operators, not a brittle proof of implementation-level reduction order.*

13 Stable design rules for future work

The following rules are worth treating as stable design commitments.

1. **Name theorems honestly.** Distinguish Dirac witnesses from real mixture theorems.
2. **Do not publish consumer surfaces before the semantics are stable.** Query syntax should follow semantic stabilization, not precede it.
3. **Prefer direct semantic queries over reuse of derived links.**
4. **Treat additive evidence as a privileged fragment, not a universal law.**
5. **Use credal or interval-valued views when hidden structure is not identifiable from observations.**
6. **Keep complete-data and observed-only semantics separate.**
7. **Reduce higher-order chains through context states unless a better semantic representation is proved.**
8. **Bridge into planning through controlled filtering, not through ad hoc symbolic chaining.**

14 The next theorem program

The most coherent next theorem program is:

1. a computable or lower-semicomputable controlled-HMM parameter class that discharges the fixed-action-stream Solomonoff hypothesis by construction;

2. stronger credal planning bounds beyond the current finite-horizon envelope theorems;

3. ambiguity-sensitive planning examples where observation-consistent model families widen the decision interval, not only the posterior interval;

4. arbitrary-Borel mixture theorems for higher-order Markov and hidden Markov settings;

5. a genuinely action-aware universal mixture, rather than the current conservative action-ignoring comparator;

6. and only after those, an HMM/POMDP query surface for symbolic consumer interfaces.

15 A compact primer for future technical conversations

Future work can safely assume the following conceptual baseline.

1. **Finite-state Markov chains:** direct path semantics is the right multi-step query surface.

2. **Higher-order Markov chains:** the right reduction is to first-order chains on context states; the sufficient statistics are $(m+1)$-gram counts.

3. **Hidden Markov models:** complete-data evidence is additive, but observed-only latent belief is posterior, not additive tally.

4. **Observed-only ambiguity:** when multiple hidden models remain compatible with the same observations, lower and upper latent support should be carried explicitly.

5. **POMDP bridge:** the structured finite-state route goes through controlled hidden Markov models and belief-state recursion.

6. **Controlled Solomonoff leg:** along fixed action streams there is now a concrete conservative universal comparator with a genuine finite-horizon regret theorem.

7. **Formalization strategy:** prefer semantic adequacy and honest boundaries over overclaimed universality or implementation mimicry.

8. **Open boundary:** the arbitrary-Borel mixture and image theorems remain the deepest unresolved measure-theoretic layer.

9. **Practical implication:** proper symbolic probabilistic reasoning requires different surfaces for additive evidence, predictive path queries, posterior latent beliefs, and ambiguity-aware credal views.

16 Conclusion

The broad picture is now reasonably clear.

The right semantics for probabilistic symbolic reasoning is not generic truth-value propagation. It is world-model semantics with query-specific evidence extraction. Additive evidence remains canonical where sufficient statistics justify it. Predictive path semantics is the right surface for Markov composition. Higher-order chains should be reduced through context states. Hidden-state models require filtering and smoothing rather than naive observation counting. Observed-only ambiguity calls for credal world-model views. Action-conditioned hidden Markov models provide the cleanest bridge to POMDP-capable world-model reasoning.

This yields a conceptually disciplined research program: one preserves the fragments that are exactly additive, one introduces posterior objects where additivity genuinely fails, one names theorems honestly, and one delays consumer surfaces until the semantic core has stabilized. That is a much cleaner foundation for future reasoning systems than either naive local probabilistic chaining or overgeneralized confidence calculus.